```
==========================================
  DAB Controller GA Optimization
  [Enhanced with Real-time Monitoring]
==========================================
>>> Scenario Loaded.
>>> Reference: Step from -5.0A to 5.0A at t=0.030s (Δ=10.0A)

>>> Starting GA Optimization...
>>> 매 세대마다 Best Individual의 성능 지표가 출력됩니다.


단일 목적 함수 최적화:
2개의 변수

Options:
CreationFcn:       @gacreationuniform
CrossoverFcn:      @crossoverscattered
SelectionFcn:      @selectionstochunif
MutationFcn:       @mutationadaptfeasible

┌──────────────────────────────────────────────────────────────────┐
│          [시험 출력] 첫 번째 개체 평가 결과                      │
├──────────────────────────────────────────────────────────────────┤
│  Parameters:  Kp =   0.1630,  Ki =     0.10                      │
│  Cost:        424130.893185  (Base: 584.3317, Penalty: 423546.5614)        │
├──────────────────────────────────────────────────────────────────┤
│  Performance Metrics (Step-Normalized):                          │
│    Step Size:       10.00 A (-5.0A → 5.0A)                       │
│    ISE:             0.548093                                     │
│    ITAE:            0.003902                                     │
│    IAE:             0.124577                                     │
│    Overshoot:       3448.50 %                                    │
│    Undershoot:      94.34 %                                      │
│    Rise Time:       6.981 ms                                     │
│    Settling Time:   70.000 ms                                    │
│    Peak Time:       37.048 ms                                    │
│    SS Error:        154.05 %                                     │
│    Peak Value:      349.846 A                                    │
│    Peak Value:      349.846 A                                    │
│    Target Value:    5.000 A                                      │
│    Control Effort:  1782.04 (Smoothness)                         │
│    Saturation:      9.01 % (Safety)                              │
│    Init Deviation:  42354.66 % (Cheat check)                     │
└──────────────────────────────────────────────────────────────────┘
>>> 이후부터는 세대(Generation)별로 Best Individual만 출력됩니다.


                                Best           Mean      Stall
Generation      Func-count      f(x)           f(x)    Generations

┌──────────────────────────────────────────────────────────────────┐
│  Generation   1                                                  │
├──────────────────────────────────────────────────────────────────┤
│  Best in Gen:  Kp =  71.6673,  Ki =  1170.46                     │
│  Gen Cost:     96.890803                                         │
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.46  (Cost: 96.8908) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174227.5
    1               40           96.89       2.191e+05        0

┌──────────────────────────────────────────────────────────────┐
|  Generation   2                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.46                 |
|  Gen Cost:     96.890803                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.46  (Cost: 96.8908) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174227.5
    2               59           96.89            2907        1

┌──────────────────────────────────────────────────────────────┐
|  Generation   3                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.46                 |
|  Gen Cost:     96.890803                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.46  (Cost: 96.8908) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174227.5
    3               78           96.89       4.832e+04        2

┌──────────────────────────────────────────────────────────────┐
|  Generation   4                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.46                 |
|  Gen Cost:     96.890803                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.46  (Cost: 96.8908) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174227.5
    4               97           96.89       4.422e+05        3

┌──────────────────────────────────────────────────────────────┐
|  Generation   5                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.46                 |
|  Gen Cost:     96.890803                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.46  (Cost: 96.8908) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174227.5
    5              116           96.89       1.693e+05        4

┌──────────────────────────────────────────────────────────────┐
|  Generation   6                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.46                 |
|  Gen Cost:     96.890803                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.46  (Cost: 96.8908) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174227.5
    6              135           96.89       3.414e+05        5

┌──────────────────────────────────────────────────────────────┐
|  Generation   7                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.43                 |
|  Gen Cost:     96.886957                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.43  (Cost: 96.8870) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   10.142 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174220.4
    7              154           96.89            1499        0

┌──────────────────────────────────────────────────────────────┐
|  Generation   8                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.39                 |
|  Gen Cost:     96.876852                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.39  (Cost: 96.8769) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   9.931 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.3
    8              173           96.88            1585        0

┌──────────────────────────────────────────────────────────────┐
|  Generation   9                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.27                 |
|  Gen Cost:     96.858115                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.27  (Cost: 96.8581) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.4
   9              192           96.86            1295        0

┌──────────────────────────────────────────────────────────────┐
|  Generation  10                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.27                 |
|  Gen Cost:     96.858115                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.27  (Cost: 96.8581) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.4
  10              211           96.86            2190        1

┌──────────────────────────────────────────────────────────────┐
|  Generation  11                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.27                 |
|  Gen Cost:     96.858115                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.27  (Cost: 96.8581) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.4
  11              230           96.86            2048        2

┌──────────────────────────────────────────────────────────────┐
|  Generation  12                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.27                 |
|  Gen Cost:     96.858115                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.27  (Cost: 96.8581) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.4
  12              249           96.86            1226        3

┌──────────────────────────────────────────────────────────────┐
|  Generation  13                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.27                 |
|  Gen Cost:     96.858115                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.27  (Cost: 96.8581) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.4
  13              268           96.86            1790        4

┌──────────────────────────────────────────────────────────────┐
|  Generation  14                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.27                 |
|  Gen Cost:     96.858115                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.27  (Cost: 96.8581) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.4
  14              287           96.86            2295        5

┌──────────────────────────────────────────────────────────────┐
|  Generation  15                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857920                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.0
  15              306           96.86            1776        0

┌──────────────────────────────────────────────────────────────┐
|  Generation  16                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857920                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.0
  16              325           96.86            1203        1

┌──────────────────────────────────────────────────────────────┐
|  Generation  17                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857920                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.0
  17              344           96.86             226        2

┌──────────────────────────────────────────────────────────────┐
|  Generation  18                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857920                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174218.0
  18              363           96.86           271.5        3

┌──────────────────────────────────────────────────────────────┐
|  Generation  19                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857867                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  19              382           96.86             226        0

┌──────────────────────────────────────────────────────────────┐
|  Generation  20                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857867                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  20              401           96.86           187.9        1

┌──────────────────────────────────────────────────────────────┐
|  Generation  21                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857867                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  21              420           96.86           187.9        2

┌──────────────────────────────────────────────────────────────┐
|  Generation  22                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857867                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  22              439           96.86           142.4        3

┌──────────────────────────────────────────────────────────────┐
|  Generation  23                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857854                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  23              458           96.86           96.86        0

┌──────────────────────────────────────────────────────────────┐
|  Generation  24                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857854                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  24              477           96.86           96.86        1

┌──────────────────────────────────────────────────────────────┐
|  Generation  25                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857854                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  25              496           96.86           96.86        2

┌──────────────────────────────────────────────────────────────┐
|  Generation  26                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857854                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  26              515           96.86           96.86        3

┌──────────────────────────────────────────────────────────────┐
|  Generation  27                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857851                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  27              534           96.86           96.86        0

┌──────────────────────────────────────────────────────────────┐
|  Generation  28                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857851                                     |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

  [Detailed Metrics]
  ➤ ISE: 0.000077  |  ITAE: 0.000025
  ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
  ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
  ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
  28              553           96.86           96.86        1

┌──────────────────────────────────────────────────────────────┐
|  Generation  29                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857851                                     |
```

```
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
   29              572           96.86           96.86        2

┌──────────────────────────────────────────────────────────────┐
|  Generation  30                                              |
├──────────────────────────────────────────────────────────────┤
|  Best in Gen:  Kp =  71.6673,  Ki =  1170.28                 |
|  Gen Cost:     96.857851                                      |
|  Best Overall: Kp =  71.6673,  Ki =  1170.28  (Cost: 96.8579) |
└──────────────────────────────────────────────────────────────┘

   [Detailed Metrics]
   ➤ ISE: 0.000077  |  ITAE: 0.000025
   ➤ Overshoot: 2.49%  |  Undershoot: 7.30%
   ➤ Rise Time: 0.150 ms |  Settling:   9.456 ms
   ➤ SS Error:  0.76%  |  Effort(dPhi): 174217.9
   30              591           96.86           96.86        3
options.MaxGenerations를 초과했기 때문에 ga이(가) 중지되었습니다.

>>> GA optimization loop finished.

============================================================
                    OPTIMIZATION COMPLETE!
============================================================
  Total Time:     57393.37 sec (956.56 min)
  Generations:    30
  Function Evals: 591
------------------------------------------------------------
  BEST PARAMETERS:
    Kp = 71.667257
    Ki = 1170.279367
  BEST COST: 96.857851
============================================================

>>> Running final verification with detailed analysis...
------------------------------------------------------------
              FINAL PERFORMANCE METRICS
          (Step-Normalized, Anti-Cheat Version)
------------------------------------------------------------
  Step Size:                       10.00 A (-5.0 → 5.0 A)
  ISE (Integral Squared Error):    0.000077
  ITAE (Integral Time Abs Error):  0.000025
  IAE (Integral Abs Error):        0.000970
  Overshoot:                       2.49 %
  Undershoot:                      7.30 %
  Rise Time (10%-90%):            0.150 ms
  Settling Time (2%):              9.456 ms
  Peak Time:                       30.775 ms
  Steady-State Error:              0.76 %
  Peak Value:                      5.249 A
  Target Value:                    5.000 A
```

```
  Init Deviation:                  11.53 % (Cheat check)
  Control Effort (Smoothness):     174217.88 (Lower is better)
  Saturation Ratio (>89deg):       13.40 % (Commercial Safety)
------------------------------------------------------------

>>> Results saved to:
    - Workspace:  GA_Result
    - MAT File:   C:\Users\vivid\Documents\Git Project\capstone\DAB-Converter-
PSIM\Simulink\Scripts\GA_Results\GA_Final_2026-01-23_20-39-37.mat
    - Metrics:    C:\Users\vivid\Documents\Git Project\capstone\DAB-Converter-
PSIM\Simulink\Scripts\GA_Results\GA_Metrics_2026-01-23_20-39-37.csv
    - Log:        C:\Users\vivid\Documents\Git Project\capstone\DAB-Converter-
PSIM\Simulink\Scripts\GA_Results\GA_Log_2026-01-23_20-39-37.txt
    - Plot:       C:\Users\vivid\Documents\Git Project\capstone\DAB-Converter-
PSIM\Simulink\Scripts\GA_Results\GA_FinalPlot_2026-01-23_20-39-37.png

>>> Optimization complete! All results and metrics saved.
>>
```